VIDYALANKAR INSTITUTE OF TECHNOLOGY

Mumbai, India

ReviewIQ

AI-Based Product Review Intelligence and Consumer Insight System

Sentiment classification with NLP, deep learning and advanced machine learning,
aspect intelligence, review trust analysis and grounded question answering

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Project Report

Submitted by

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Certificate

This is to certify that the project report entitled **"ReviewIQ: AI-Based Product Review Intelligence and Consumer Insight System"** is a bonafide record of the work carried out by

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at **Vidyalankar Institute of Technology** under my guidance, and is submitted in partial fulfilment of the requirements of the project course.

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Declaration

We declare that this written submission represents our ideas in our own words and that, where others' ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented, fabricated or falsified any idea, data, fact or source in our submission.

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Abstract

Online product reviews are abundant but unstructured, and a single popular product can attract thousands of opinions that no person can read in full. This project, **ReviewIQ**, turns raw Amazon Electronics reviews into decisions for two audiences: buyers who ask "should I buy this?" and sellers who ask "what should we improve?".

The system combines three families of techniques on one shared, metadata-enriched corpus of 50,000 reviews covering 34,365 products. The **NLP** branch uses a VADER lexicon baseline, TF-IDF with logistic regression and a linear SVM, LDA topic modelling and aspect-based sentiment analysis. The **deep learning** branch trains a bidirectional LSTM with attention from scratch and fine-tunes DistilBERT. The **advanced machine learning** branch predicts review helpfulness with tuned XGBoost, clusters reviews with MiniBatchKMeans and flags potentially suspicious reviews with an unsupervised Isolation Forest. A semantic search and retrieval-augmented question-answering layer lets users ask product-specific questions and always shows the supporting reviews. Everything is delivered through a nine-page interactive Streamlit dashboard with a live sentiment classifier.

On a shared test set of 10,000 reviews, the BiLSTM reaches 88.8% accuracy and the highest macro-F1 (0.718), significantly above a TF-IDF SVM, while DistilBERT reaches the highest accuracy (89.9%) using only 6,000 training reviews. A three-seed ablation shows that class weighting, not attention, accounts for most of the LSTM's macro-F1 gain, and that an unweighted LSTM matches DistilBERT's accuracy. The LSTM is about 57 times faster than DistilBERT on a CPU with 26 times fewer parameters. Battery is the most criticised aspect (20.2% negative mentions). A similarity-floor guardrail makes the question-answering component decline 95.6% of plausible but unanswerable questions and 100% of off-topic ones, while accepting 97.8% of answerable questions. Limitations, including weak labels from star ratings and a non-random sample, are stated explicitly.

Keywords: sentiment analysis, LSTM, attention, DistilBERT, aspect-based sentiment analysis, XGBoost, Isolation Forest, retrieval-augmented generation, product reviews.

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List of Abbreviations

ABSA	Aspect-Based Sentiment Analysis
AML	Advanced Machine Learning
ASIN	Amazon Standard Identification Number
BiLSTM	Bidirectional Long Short-Term Memory
CI	Confidence Interval
DL	Deep Learning
LDA	Latent Dirichlet Allocation
LSTM	Long Short-Term Memory
NLP	Natural Language Processing
RAG	Retrieval-Augmented Generation
SVM	Support Vector Machine
TF-IDF	Term Frequency, Inverse Document Frequency
VADER	Valence Aware Dictionary and sEntiment Reasoner
XGBoost	Extreme Gradient Boosting

Chapter 1

Introduction

1.1 Background and Motivation

Customer reviews influence purchasing decisions and guide product development. On a large marketplace, however, a popular product can have thousands of free-text opinions, and the star average hides what customers actually say. A buyer wants to know what people like and dislike about *this* product. A seller wants to know which problems recur across products and which reviews are worth reading. Reading everything is impossible, so automated review analysis is central to modern e-commerce.

The problems involved are usually studied separately: sentiment classification, aspect extraction, helpfulness prediction and opinion-spam detection often use different datasets and evaluation protocols. It is therefore hard to judge what each family of methods (lexicon and feature-based NLP, neural sequence models, ensemble and unsupervised learning) contributes to a practical system.

1.2 Problem Statement

To design, implement and evaluate an end-to-end system that classifies customer review sentiment into positive, neutral and negative classes, extracts aspect-level insight, assesses review quality and trust, and answers natural-language questions about a product using only its own reviews, serving both buyers and businesses through one interactive interface.

1.3 Objectives

1. Prepare a clean, labelled review corpus enriched with product names, brands and images.
2. Classify review sentiment with models from three families (NLP, deep learning and advanced ML) and compare them fairly on one test set.
3. Build an LSTM-based sentiment classifier, report its performance metrics and visualise the language of each sentiment with word clouds.
4. Derive aspect-level sentiment, topics and clusters to show what customers talk about and complain about.

5. Predict review helpfulness and flag potentially suspicious reviews.
6. Provide semantic search and a grounded question-answering interface that shows its evidence and declines unsupported questions.
7. Deliver all results through a dashboard for buyers, businesses and model inspection.

1.4 Scope

The project covers one product category (Amazon Electronics), English-language reviews, and a 50,000-review working sample. Sentiment labels are derived from star ratings. Suspicious-review detection is unsupervised and makes no claim of identifying fake reviews with a measured accuracy.

1.5 Contributions

- A three-pillar framework integrating NLP, deep learning and advanced machine learning over a shared corpus.
- A like-for-like comparison of five sentiment models with bootstrap confidence intervals and paired significance tests.
- A three-seed ablation separating the effect of attention pooling from that of class weighting, and a measured cost comparison.
- Product-level intelligence: aspect complaint ranking, helpfulness prediction and a corrected unsupervised suspicious-review score.
- Grounded question answering with a similarity floor whose effect is measured.
- A candid account of limitations and of the engineering problems found and fixed during development.

1.6 Organisation of the Report

Chapter 2 surveys related work. Chapter 3 presents the analysis and design. Chapter 4 describes the dataset. Chapter 5 explains the implementation. Chapter 6 reports testing and results. Chapter 7 concludes and lists future work. Appendices show dashboard screenshots and how to run the project.

Chapter 2

Literature Survey

2.1 Sentiment Classification

Early work framed review sentiment as supervised classification with bag-of-words features [1], where linear support vector machines remain a strong baseline [6]. Lexicon-and-rule systems such as VADER [7] need no training data and are a common zero-shot baseline. Recurrent networks, notably the LSTM [8], model word order and negation, and attention mechanisms [9] let a classifier weight the most informative tokens; attention-based LSTMs were applied to aspect-level sentiment by Wang et al. [10]. Pre-trained transformers [11], including the distilled DistilBERT [12], now dominate benchmarks.

2.2 Aspects, Topics and Review Quality

Feature-level opinion mining [2] extracts product aspects and the sentiment expressed towards them, and topic models such as LDA [13] discover themes without labels. Helpfulness has been predicted from structural, lexical and metadata features [3], which suits gradient-boosted trees [14]. Opinion spam is hard to study because labelled ground truth is scarce [4, 5]; unsupervised outlier detection such as Isolation Forest [15] is a pragmatic option when no labels exist.

2.3 Retrieval and Grounded Generation

Sentence encoders [16, 17] retrieve by meaning rather than keyword overlap, and retrieval-augmented generation [18] conditions a generator on retrieved passages so that answers can be traced to evidence. Instruction-tuned encoder-decoder models [19, 20] are small enough to run locally. A known weakness is that generators may answer confidently even when retrieval is weak.

2.4 Summary of Gaps Addressed

Table 2.1: Gaps in typical review-analysis work and how this project addresses them.

Typical gap	How ReviewIQ addresses it
Tasks studied in isolation	One corpus feeds sentiment, aspects, helpfulness, trust and question answering
Single models reported as point estimates	Five models on one test split with bootstrap intervals and paired tests
Product identified by cryptic ID	Product names, brands and images joined via the parent ASIN
Opinion-spam claims without ground truth	Unsupervised score with explicit wording and no accuracy claim
Generated answers without evidence	Retrieved reviews always shown; similarity floor declines unsupported questions

Chapter 3

System Analysis and Design

3.1 Requirements

3.1.1 Functional requirements

- Load, clean and label a review corpus and join product metadata.
- Classify the sentiment of any review text with several models and report metrics.
- Show, per product, a summary, aspect sentiment with supporting sentences, filterable reviews and review-trust analysis.
- Compare two products using computed statistics only.
- Answer product-specific questions from that product's reviews and show the evidence.
- Provide corpus-wide business views: complaint ranking, topics, clusters, helpfulness and suspicious reviews.
- Provide a model-inspection area with metrics, training curves, word clouds and a live classifier.

3.1.2 Non-functional requirements

Responsiveness (pages must render in seconds, with heavy computation cached), explainability (every insight links to evidence), honesty (limitations and uncertainty stated), no login or personal data, and reproducibility (every reported number regenerated by a script).

3.2 System Architecture

Figure 3.1 shows the architecture. Cleaned and labelled reviews feed three parallel branches whose outputs are fused into product-level intelligence and a semantic search and retrieval-augmented question-answering layer, all exposed through one dashboard.

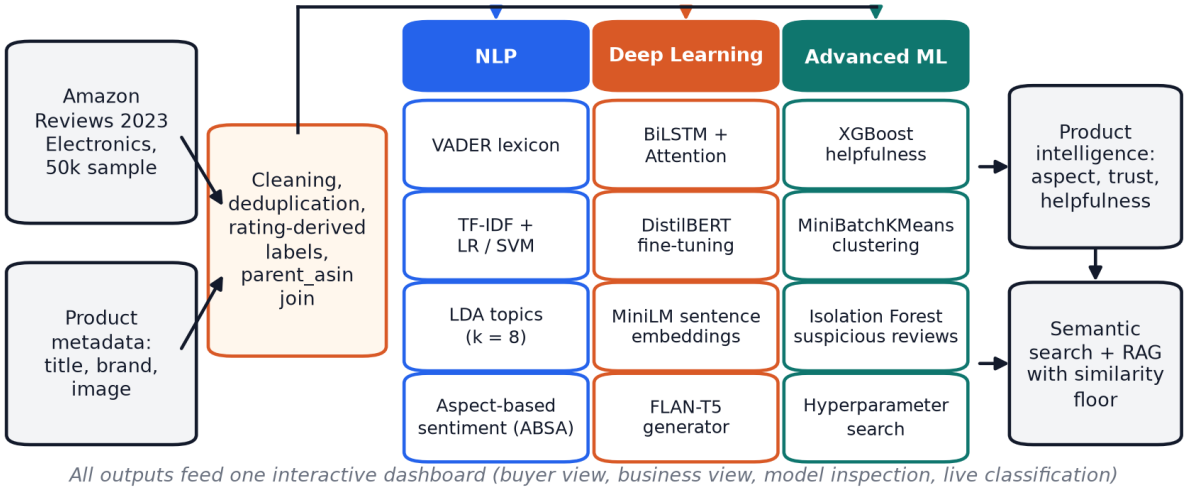


Figure 3.1: Overall architecture of ReviewIQ.

3.3 Data Flow

1. Raw review records are read, cleaned, de-duplicated and labelled (positive, neutral, negative from star ratings), producing the processed review table.
2. Product metadata (title, brand, image) is fetched and joined on the parent ASIN, producing the products table. Reviews are never duplicated per product attribute.
3. Each branch reads the processed table and writes its own artefacts: models, prediction tables, metrics reports and figures.
4. All reviews are encoded once into sentence embeddings that are reused by semantic search, the question-answering layer and the suspicious-review feature that measures near-duplicate text.
5. The dashboard loads the tables and artefacts, joins them by product, and renders each page. Heavy computations are cached.

3.4 Module Description

3.5 Technology Stack

3.6 Environment

Training and evaluation were run on a laptop with an NVIDIA GeForce RTX 4050 Laptop GPU (6 GB) and an 8-thread CPU under Linux (WSL2). DistilBERT was fine-tuned on CPU.

Table 3.1: Modules of the system.

Module	Responsibility
Data preparation	Cleaning, de-duplication, sampling, rating-derived labels, metadata join
NLP	VADER, TF-IDF with logistic regression and SVM, LDA topics, aspect-based sentiment
Deep learning	BiLSTM with attention, fine-tuned DistilBERT, word clouds, live classifier
Advanced ML	XGBoost helpfulness, MiniBatchKMeans clustering, Isolation Forest suspicious reviews
Semantic search and RAG	Sentence embeddings, product-scoped retrieval, FLAN-T5 answers, similarity floor
Dashboard	Nine Streamlit pages for buyers, businesses and model inspection
Evaluation	Shared-split metrics, bootstrap intervals, ablation, cost, retrieval and guardrail experiments

Table 3.2: Technologies used.

Purpose	Tools
Language and data	Python 3.12, pandas, NumPy
Classical ML	scikit-learn 1.9, XGBoost 3.4
Deep learning	PyTorch 2.13 (CUDA), Hugging Face Transformers 5.16
Sentiment baseline	vaderSentiment
Embeddings and retrieval	Sentence-Transformers (all-MiniLM-L6-v2)
Generation	FLAN-T5-base
Visualisation	Matplotlib, wordcloud
User interface	Streamlit 1.62 with custom CSS
Testing and capture	Streamlit AppTest, Playwright
Documents	LaTeX (Tectonic), python-pptx

Chapter 4

Dataset and Preprocessing

4.1 Dataset

The Electronics category of the Amazon Reviews 2023 collection (McAuley Lab) [21] was used. A working sample of 50,000 reviews was prepared. Table 4.1 summarises it.

Table 4.1: Statistics of the 50,000-review working sample.

Property	Value
Reviews / products / reviewers	50,000 / 34,365 / 8,883
Review dates	2000 to March 2023
Mean / median length (words)	71.7 / 39
Mean star rating	4.25
Verified purchases	80.2%
Reviews with at least one helpful vote	26.2%
Positive / neutral / negative	40,205 / 3,506 / 6,289
share of corpus	80.4% / 7.0% / 12.6%
Products matched to metadata	34,364 / 34,365

4.2 Cleaning Steps

Records without a rating, or with fewer than three words after cleaning, were dropped. HTML entities, tags and URLs were removed. Duplicate (reviewer, product, text) triples were removed. The review title and body were concatenated into one text field. The sample was stratified by sentiment class and shuffled with a fixed seed.

4.3 Labels

Sentiment labels are weakly supervised from star ratings: ratings of 4 and 5 are positive, 3 is neutral, and 1 and 2 are negative. This is standard practice at scale but introduces label noise: a three-star review can be warm, and a five-star review can contain a complaint. Because the classes are strongly imbalanced, macro-averaged F1 is used as the primary metric.

4.4 Product Metadata Integration

Reviews carry only a product identifier. The category's metadata file (about 5.2 GB) was streamed and filtered to the parent ASINs present in the sample. Title, brand, image and category were joined on the parent ASIN, matching 34,364 of 34,365 products, and stored in a separate products table. Grouping by parent ASIN (the product listing) rather than the variant-level ASIN means colour and size variants roll up into one product.

4.5 Sampling Caveat

The working sample was taken from the beginning of the category file rather than drawn at random. That file is grouped by reviewer (the 65,552 raw records form 9,686 contiguous blocks, one per reviewer), so the sample over-represents prolific reviewers: 392 accounts with at least 20 reviews contribute about 40% of it. This is discussed again where it affects the suspicious-review feature.

Chapter 5

Implementation

5.1 NLP Branch

5.1.1 VADER baseline

VADER produces a compound score between -1 and 1. Scores of at least 0.05 are labelled positive, at most -0.05 negative, and the rest neutral. It needs no training and serves as the zero-shot baseline.

5.1.2 TF-IDF with logistic regression and linear SVM

Reviews are vectorised with TF-IDF using unigrams and bigrams, 30,000 features, minimum document frequency 2 and sublinear term frequency. A multinomial logistic regression and a linear SVM are trained with balanced class weights on the 40,000-review training split.

5.1.3 LDA topic modelling

LDA with eight topics is fit with online variational inference on a count matrix of 3,000 unigram and bigram features (stop words removed), using an 8,000-document subsample for fitting and applying the model to all reviews. Table 5.1 lists the topics.

Table 5.1: LDA topics and their leading words.

#	Interpretation	Top words
0	Audio and speakers	sound, speaker, music, speakers, audio, bluetooth, quality, volume
1	Storage cards, Apple accessories	card, ipad, cord, apple, sd, charging, sd card, phone, waste
2	General praise, computers, drives	great, works, works great, drive, price, screen, computer
3	USB and power accessories	usb, power, laptop, mouse, nice, good, work
4	Cameras, TVs, mounts	camera, tv, light, video, mount, easy
5	Cases and tablets	case, fit, perfect, love, tablet, keyboard
6	Headphones and phones	great, sound, phone, quality, headphones, work, easy
7	Cables and networking	cable, cables, router, wifi, hdmi, devices, signal

5.1.4 Aspect-based sentiment analysis

Ten aspects (battery, camera, display, sound, performance, build quality, price and value, connectivity, size and comfort, customer service) are defined by keyword lexicons. Each review is split into sentences; sentences containing an aspect keyword are scored with VADER, and the aspect receives a positive, neutral or negative polarity. The matched sentences are kept as human-readable evidence, so every aspect judgement in the interface can be inspected.

5.2 Deep Learning Branch

5.2.1 BiLSTM with attention

Tokens (lower-cased alphanumerics plus "!" and "?") are mapped to a vocabulary built on the training split only (14,937 words, minimum frequency 3), and sequences are truncated to 200 tokens (Figure 5.1). A learned 128-dimensional embedding feeds a two-layer bidirectional LSTM with 128 units per direction. With hidden states $\mathbf{h}_t$, attention pooling computes

$$\alpha_t = \frac{\exp(\mathbf{w}^\top \mathbf{h}_t)}{\sum_s \exp(\mathbf{w}^\top \mathbf{h}_s)}, \qquad \mathbf{c} = \sum_t \alpha_t \mathbf{h}_t, \tag{5.1}$$

with padding masked, and $\mathbf{c}$ passes through dropout and a linear layer to a three-way softmax. Because the classes are imbalanced, the loss uses class weights $w_c = \sqrt{N/(3n_c)}$, and the best epoch is chosen by validation macro-F1 rather than loss.

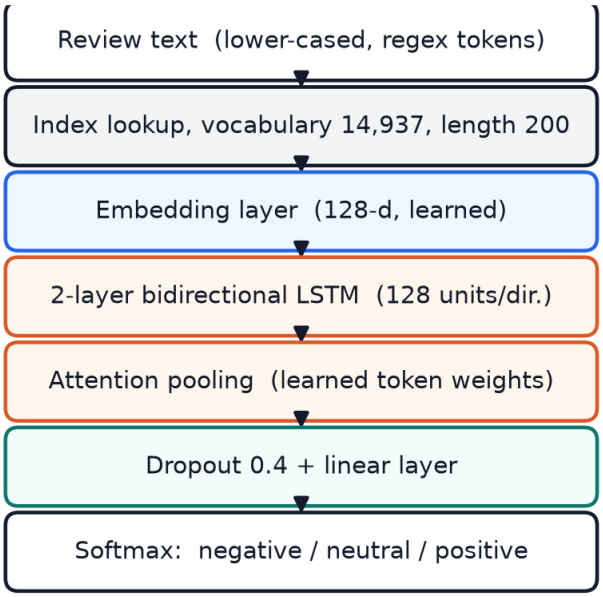


Figure 5.1: Architecture of the BiLSTM with attention pooling.

Table 5.2: BiLSTM configuration.

Setting	Value
Vocabulary / maximum length	14,937 / 200 tokens
Embedding / hidden size / layers	128 / 128 per direction / 2
Trainable parameters	2,572,420
Optimiser	Adam, learning rate 2×10^{-3}, weight decay 10^{-5}
Batch size / gradient clipping	128 / 1.0
Dropout	0.4
Epochs / early stopping	at most 12 / patience 3 on validation macro-F1
Training / validation / test reviews	36,000 / 4,000 / 10,000
Training time	about 7 minutes on a laptop GPU

5.2.2 Fine-tuned DistilBERT

The pre-trained distilled transformer is fine-tuned with a three-way classification head (maximum length 128, AdamW optimiser, learning rate 2×10^{-5}, 2 epochs, batch size 8) on a stratified 6,000-review subsample, because the run was performed on a CPU (about 60 minutes).

5.2.3 Word clouds and live classifier

Word clouds are drawn per sentiment class, weighting each word by its within-class frequency multiplied by its over-representation relative to the whole corpus, so that distinctive rather than merely frequent vocabulary is shown. The dashboard's live classifier runs the LSTM, DistilBERT and the SVM side by side on any text and highlights the words the LSTM attended to.

5.3 Advanced Machine Learning Branch

5.3.1 XGBoost helpfulness prediction

The target is whether a review received at least one helpful vote. Ten engineered features are used (Table 5.3). The model is tuned with 15 randomised-search candidates under 3-fold cross-validation optimising F1, with the positive-class weight set for the imbalance.

5.3.2 MiniBatchKMeans clustering

Six clusters are computed on a separate TF-IDF representation (5,000 features, stop words removed) so that function words do not dominate.

Table 5.3: Features of the helpfulness model.

Feature	Meaning
rating	Star rating
review_length_words, title_length_words	Length of the review body and title
verified_purchase	Verified-purchase flag
exclamation_count, question_count	Count of "!" and "?"
capital_ratio	Share of capital letters
vader_compound	VADER sentiment of the text
review_age_days	Age relative to the newest review
num_aspects_mentioned	Distinct aspects found by the ABSA module

5.3.3 Isolation Forest for suspicious reviews

With no labelled fake or genuine reviews, suspicious-review detection is framed as unsupervised anomaly detection. Features are a binary indicator for very short reviews, capital-letter ratio, exclamation count, verified-purchase flag, the logarithm of the number of reviews the same account has in the sample, and the maximum cosine similarity to another review of the same product (computed per product on sentence embeddings). After standardisation, an Isolation Forest with contamination 0.10 produces an anomaly score rescaled to $[0, 1]$. A rule-based reason string accompanies each flag. The interface calls flagged reviews only "potentially suspicious based on automated pattern analysis".

5.4 Semantic Search and Question Answering

All reviews are encoded once with a MiniLM sentence encoder into 384-dimensional unit vectors, so cosine similarity is a dot product. For a question about a product, retrieval is restricted to that product's reviews and the top five by cosine similarity are returned. FLAN-T5-base is prompted with the question and the retrieved text (each review truncated to 300 characters) and asked to answer in one or two sentences using only that text. Small generators do not reliably follow "say you do not know" instructions, so a hard *similarity floor* is applied: if the best similarity is below 0.30, the system answers that the reviews do not provide enough information and does not call the generator. The retrieved reviews and similarities are always displayed.

5.5 Dashboard

The dashboard is a nine-page Streamlit application, styled with custom CSS and organised by a sidebar (Table 5.4). Summaries and product comparisons are generated from computed statistics by templates, never by a language model, so they cannot introduce unsupported claims.

Screenshots are in Appendix A.

Table 5.4: Dashboard pages.

Page	Content
Home	Project overview, pipeline and the three pillars (NLP, deep learning, advanced ML)
Explore Products	Search and browse products by name, with images, rating and review count
Product Insights	Overview and AI summary; aspect analysis with evidence sentences; filterable reviews with LSTM predictions; review trust table with threshold slider
Compare Products	Two products side by side with a template-generated comparison
Ask ReviewIQ	Product-scoped question answering with evidence cards
Business Intelligence	Scope selector; complaint ranking; topics and clusters; helpfulness and suspicious reviews
Deep Learning Lab	LSTM architecture, training curves, metrics, five-model comparison, word clouds, live classifier
Model Insights	NLP, deep learning and advanced ML tabs with full reports
About	Dataset, pipeline and limitations

5.6 Engineering Problems Found and Fixed

Several problems were found by testing and inspection during development. They are recorded here because the fixes shaped the final design.

1. **Backwards suspicious-review features.** An early version fed raw rating and raw review length into the Isolation Forest. The scores correlated $+0.55$ with length and -0.42 with rating, so long, detailed, negative reviews were flagged as suspicious simply because they are rare. Replacing length by a "very short" indicator, removing rating and adding per-account review volume reversed this (correlations $+0.06$ and $+0.25$ with length and account volume).

2. **Ungrounded answers.** The unguarded generator answered "No" to a warranty question that the reviews never mention, at a top similarity of about 0.25. The similarity floor was added as a result.

3. **Slow, stuck pages.** The helpfulness view recomputed VADER features for all 50,000 reviews on every interaction (about 23 seconds), leaving the page dimmed. Caching the feature-building step and the model removed the delay.

4. **Missing product images.** Twenty products stored their image under a field name the parser did not read, and the fallback image URL returned an access error. The parser was corrected (four products remain without any image in the source data) and the fallback became a built-in image.

5. **Ignored sidebar clicks.** The sidebar radio's identity changed with every page, so Streamlit dropped the first click after most page changes. A stable widget key with callback-based buttons fixed it; all twelve tested transitions and both in-page navigation buttons now work on the first click.
6. **Duplicate navigation.** Streamlit treated a folder named `pages` as its own multipage navigation and showed a second menu. The folder was renamed to `sections`.

Chapter 6

Testing and Results

6.1 Evaluation Protocol

All sentiment models are evaluated on one stratified 80/20 split (seed 42), giving 40,000 training and 10,000 test reviews. For the LSTM, 10% of the training portion is held out for early stopping, and the test set is never used for model selection. VADER needs no training and is scored on the same 10,000 test reviews. Accuracy and macro-F1 are reported with 95% percentile bootstrap confidence intervals (2,000 resamples of the test set) and with paired bootstrap differences between models.

6.2 Sentiment Classification

Table 6.1: Sentiment results on the shared 10,000-review test set. Accuracy and macro-F1 show the half-width of the 95% bootstrap interval; best per column in bold.

Model	Accuracy	Macro-F1	$F1_{neg}$	$F1_{neu}$	$F1_{pos}$
VADER	.813±.008	.512±.012	.551	.080	.906
TF-IDF + LR	.860±.007	.689±.013	.733	.400	.935
TF-IDF + SVM	.888±.006	.693±.014	.747	.382	.950
BiLSTM + Attention	.888±.006	**.718**±.013	.754	**.446**	.954
DistilBERT	**.899**±.006	.702±.015	**.780**	.368	**.958**

Every trained model clearly outperforms the lexicon baseline, whose neutral-class F1 is only 0.08. The two neural models are strongest, but in different ways: DistilBERT has the highest accuracy and negative and positive F1, while the BiLSTM has the highest macro-F1 because it recovers the neutral class better. The BiLSTM's macro-F1 advantage over the SVM

Table 6.2: Paired bootstrap differences (first minus second) with 95% intervals. Intervals excluding zero are significant.

Comparison	Δ Macro-F1 [CI]	Δ Accuracy [CI]
BiLSTM − SVM	+.025 [+.010, +.039]	+.001 [−.005, +.006]
BiLSTM − DistilBERT	+.016 [−.000, +.031]	−.011 [−.016, −.005]
DistilBERT − SVM	+.009 [−.008, +.026]	+.011 [+.005, +.017]

is significant, its advantage over DistilBERT is borderline and not significant at the 95% level,

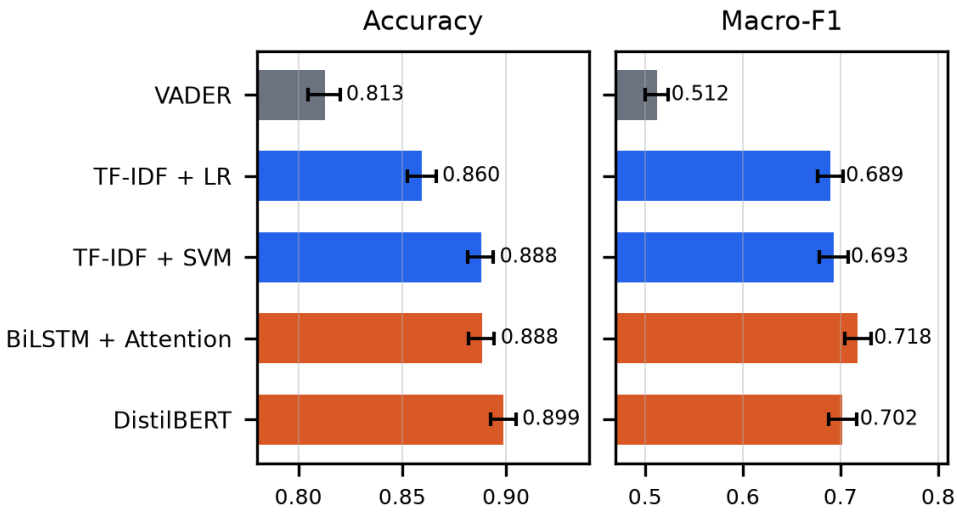


Figure 6.1: Accuracy and macro-F1 with 95% bootstrap confidence intervals.

and DistilBERT's accuracy advantage over both is significant. The BiLSTM and the SVM are indistinguishable in accuracy; the neural model's gain is concentrated on minority classes.

6.3 BiLSTM Training Behaviour

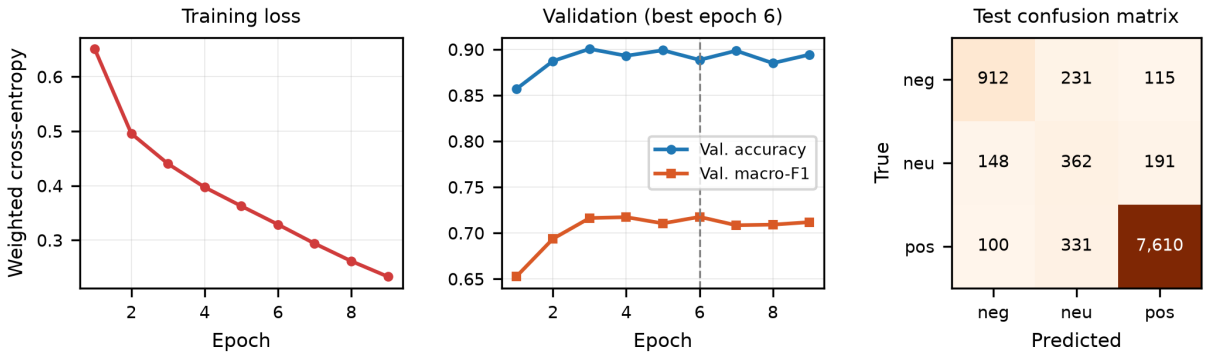


Figure 6.2: BiLSTM training loss, validation metrics and test confusion matrix.

Training loss falls steadily, while validation macro-F1 plateaus after epoch three and peaks at epoch six, after which the validation metrics stop improving and early stopping ends training. Most errors involve the neutral class: 191 of the 701 neutral test reviews are predicted positive and 148 negative, while 331 of the 8,041 positive reviews are predicted neutral.

6.4 Ablation: What Drives the BiLSTM's Gain?

Three variants were trained with three seeds each on the identical split: the full model, the same model with an unweighted loss, and a mean-pooling variant that replaces attention (weighted loss retained). Class weighting is the dominant factor. Removing it lowers macro-F1 by 0.012

Table 6.3: LSTM ablation (mean ± standard deviation over three seeds). The DistilBERT row is a single run.

Variant	Accuracy	Macro-F1	$F1_{neu}$
Full (attention + class weights)	.891±.003	**.722**±.004	**.445**±.000
No class weights	**.899**±.001	.709±.008	.405±.013
Mean pooling (class weights)	.889±.003	.715±.005	.430±.004
DistilBERT (reference)	.899	.702	.368

on average (in all three seeds) and lowers neutral F1 from 0.445 to 0.405, but raises accuracy by 0.007 in every seed. Replacing attention with mean pooling lowers macro-F1 by 0.007 on average and leaves accuracy unchanged within noise, so attention contributes a small and less consistent improvement. The seed-to-seed standard deviation of the full model (about 0.004) is much smaller than its gap to the SVM (about 0.03). An LSTM trained without class weights matches DistilBERT's accuracy (0.899) while still exceeding its macro-F1 on average (0.709 versus 0.702), which suggests DistilBERT's accuracy lead reflects the accuracy and macro-F1 trade-off more than a clear difference in capacity.

6.5 Computational Cost

Table 6.4: Model size and inference throughput (reviews per second).

Model	Parameters	CPU	GPU
TF-IDF + SVM	90 k	13,550	n/a
BiLSTM + Attention	2.57 M	975	5,688
DistilBERT	66.96 M	17	451

The BiLSTM has 26 times fewer parameters than DistilBERT and classifies about 57 times faster on CPU and 13 times faster on GPU. For batch scoring or CPU-only serving it offers a much better balance of cost and accuracy; the SVM is fastest but has the lowest macro-F1 of the three.

6.6 Aspect-Level Results

The lexicon matched 64,134 aspect mentions in 33,257 reviews (66.5%). Battery is the most criticised aspect at 20.2% negative mentions.

Table 6.5: Aspect-level sentiment (% of mentions), ranked by negative share.

Aspect	Positive	Neutral	Negative	Mentions
Battery	56.3	23.5	20.2	6,968
Connectivity	56.7	23.9	19.3	4,166
Customer service	64.0	16.7	19.3	2,958
Performance	58.7	25.5	15.8	4,726
Display	64.4	21.1	14.5	4,540
Sound	73.4	13.7	12.9	8,327
Camera	70.1	18.6	11.3	5,991
Price and value	74.2	15.1	10.7	11,070
Size and comfort	76.1	13.5	10.4	10,752
Build quality	66.5	25.4	8.1	4,636

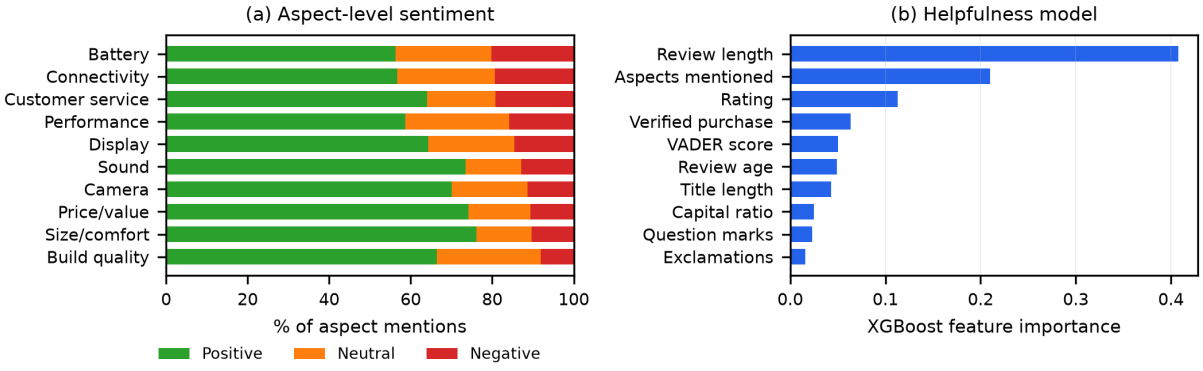


Figure 6.3: (a) Aspect-level sentiment shares. (b) XGBoost feature importance.

6.7 Topics and Clusters

The LDA topics (Table 5.1) separate along product families. Clustering is less informative (Table 6.6): only two of six KMeans clusters are clearly topical, and the other four are dominated by generic praise. This is reported as a negative finding: on short, rating-skewed reviews, TF-IDF clustering mostly recovers sentiment style rather than subject, and LDA is the better tool for topic discovery.

Table 6.6: KMeans clusters (k = 6).

Cluster	Reviews	Mean rating	Leading terms
0	19,309	3.97	use, work, easy, camera, great, works, usb
1	3,570	4.80	works, works great, great, stars, works perfectly
2	13,552	4.24	stars, great, good, perfect, price, case, nice
3	3,228	4.80	product, love, great product, good product
4	5,101	4.52	laptop, good, fits, nice, bag, case
5	5,240	4.30	sound, quality, headphones, speaker, price, ear

6.8 Helpfulness Prediction

The tuned XGBoost model (200 trees, depth 6, learning rate 0.01, subsample 0.7, column sample 0.7) reaches 70.5% accuracy with helpful-class precision 0.458, recall 0.702 and F1 0.555 (confusion matrix: 5,211 and 2,172 for the unhelpful class; 779 and 1,838 for the helpful class). Accuracy is modest because helpfulness is a noisy target that plausibly depends on exposure, which no review-level feature captures. Review length is the most important feature, followed by the number of aspects mentioned and the rating.

6.9 Suspicious-Review Detection

There is no ground truth, so no accuracy is reported. The Isolation Forest flags 4,999 reviews (10.0%), of which 36 are near-duplicates (similarity at least 0.95 within a product). The highest-scoring reviews are templated one-liners such as “Five Stars. Love it!!!”, all-capitals emotional text, and effusive praise from high-volume unverified accounts. The dashboard exposes a threshold slider and a table showing the flag, score, rating, review text and the signals behind each flag.

6.10 Retrieval and the Similarity Floor

Ten paraphrased queries, one per aspect and deliberately avoiding every lexicon keyword, were used. Dense retrieval achieved a mean aspect-mention rate at five of 0.54 against 0.32 for TF-IDF retrieval. It was better on seven aspects, tied on one, worse on display, and both methods scored zero on the battery query. With ten queries the difference is indicative, not conclusive. The guardrail declines all off-topic and 95.6% of plausible but unanswerable questions while

Table 6.7: Retrieval and guardrail results.

(a) Aspect-mention rate at five (10 paraphrased queries)		
Method	Mean	Queries at least 0.6
TF-IDF cosine	0.32	3 of 10
MiniLM dense	0.54	6 of 10
(b) Top similarity (9 products, 5 questions per group)		
Question group	Mean	Below the 0.30 floor
Answerable	0.455	2.2%
Plausible but unanswerable	0.218	95.6%
Off-topic	0.184	100%

declining only 2.2% of answerable ones. The faithfulness of answers that pass the floor was not measured.

6.11 Word Clouds

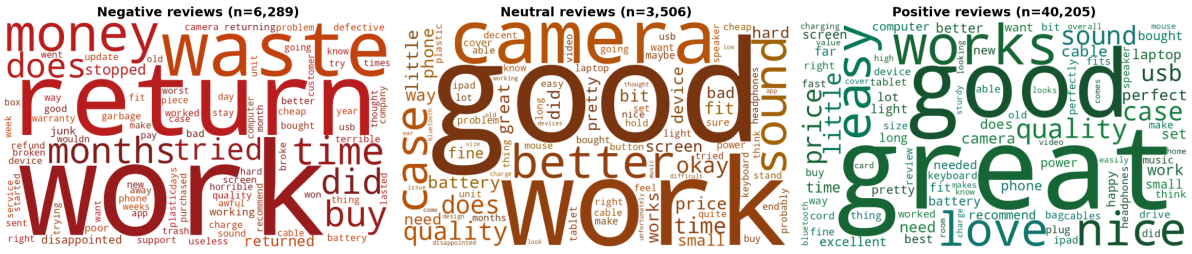


Figure 6.4: Distinctive-vocabulary word clouds for negative, neutral and positive reviews.

Negative reviews are dominated by return and failure language (“return”, “waste”, “money”, “stopped”, “broke”), positive reviews by ease and satisfaction (“easy”, “love”, “nice”, “perfect”), and neutral reviews by hedged evaluation (“okay”, “fine”, “better”) mixed with product nouns, consistent with the difficulty of the neutral class.

6.12 Software Testing

- **Page rendering:** every one of the nine dashboard pages was executed headlessly with Streamlit's testing framework and produced no exceptions.
- **Navigation:** twelve consecutive sidebar transitions and both in-page navigation buttons were exercised in a real browser; all work on the first click after the fix described in Section 5.6.
- **Visual checks:** screenshots of every page were reviewed for clipping, contrast and missing images.
- **Guardrail and isolation:** questions scoped to one product never returned another product's reviews as evidence, and unsupported questions were declined.
- **Reproducibility:** each reported number is regenerated from saved artefacts by a script, with fixed seeds.

Chapter 7

Conclusion and Future Scope

7.1 Conclusion

ReviewIQ integrates NLP, deep learning and advanced machine learning around a shared review corpus and exposes the results through a grounded, product-scoped dashboard. A BiLSTM with attention trained from scratch matched a TF-IDF SVM in accuracy but significantly improved macro-F1 by better recognising the neutral class, while a lightly fine-tuned DistilBERT gave the best accuracy of the default-trained models; the two neural models were statistically comparable on macro-F1. A three-seed ablation showed that class weighting, rather than attention, accounts for most of the LSTM's macro-F1 gain, and that an unweighted LSTM matches DistilBERT's accuracy. The LSTM runs about 57 times faster than DistilBERT on a CPU with 26 times fewer parameters. Aspect analysis identified battery, connectivity and customer service as the main complaint areas, helpfulness could be predicted moderately well from review structure, and an unsupervised score surfaced low-effort and high-volume review patterns once its features were corrected. A similarity floor made the question-answering component decline nearly all unsupported questions.

7.2 Limitations

- Sentiment is derived from stars, not human annotation; reported accuracies measure agreement with ratings.
- The sample is a contiguous block of a reviewer-grouped file and covers one English-language category, so it over-represents prolific reviewers and may not transfer to other domains.
- DistilBERT was fine-tuned on one-sixth of the data used for the LSTM and without class weighting; the main comparison uses single runs, although the ablation shows small seed variance for the LSTM.
- The aspect module uses keyword lexicons and sentence-level VADER scores, so it misses implicit aspects and can mis-score sarcasm.
- Flagged reviews are anomalies, not confirmed fakes, and coordinated campaigns that imitate normal behaviour would not be detected.

- The faithfulness and fluency of generated answers were not measured.

7.3 Future Scope

- Train DistilBERT on the full data with matched class weighting for a fairer comparison.
- Replace keyword aspects with learned aspect extraction.
- Collect human annotations for neutral and suspicious subsets to measure real accuracy.
- Draw a random sample across several categories and languages.
- Evaluate the faithfulness of generated answers and try larger generators.
- Add multi-product dashboards for sellers and an automated alert when a new complaint area emerges.

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Appendix A

Dashboard Screenshots

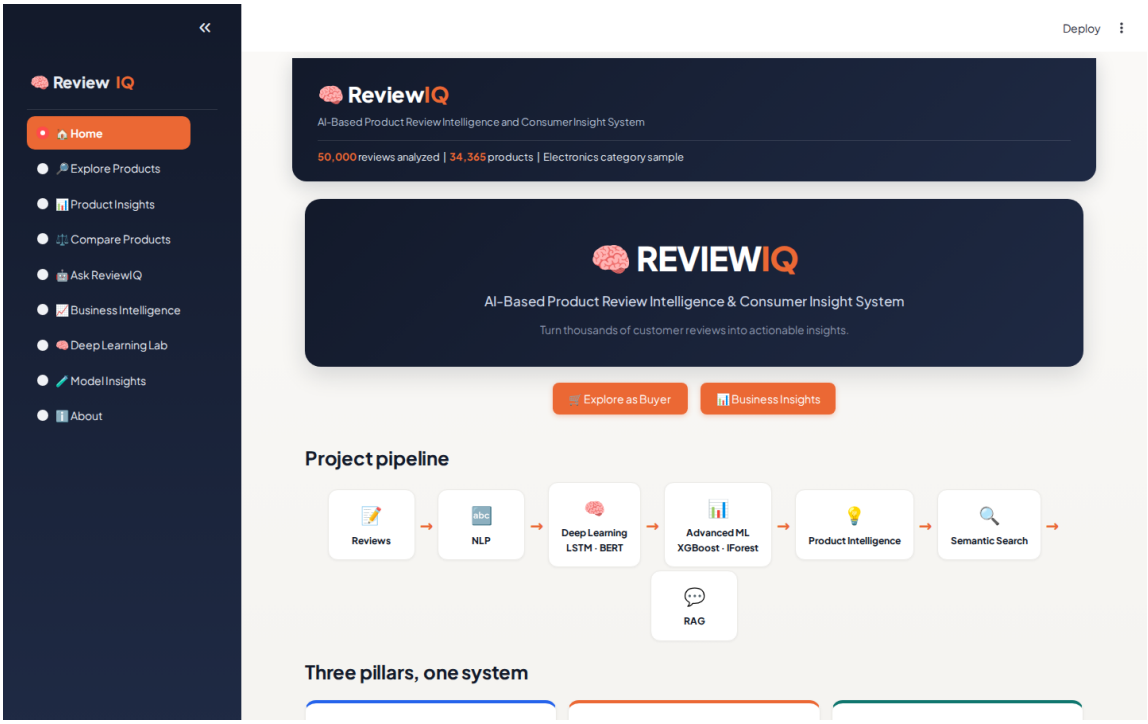


Figure A.1: Home page.

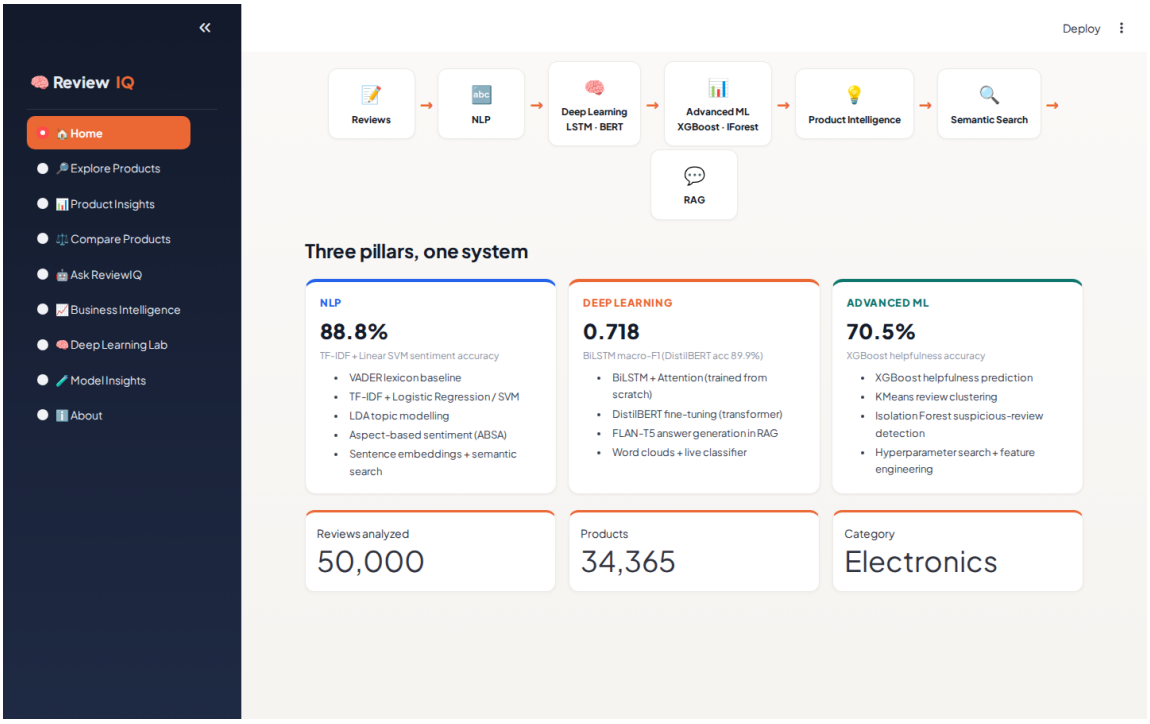


Figure A.2: Home page: pipeline and the three pillars.

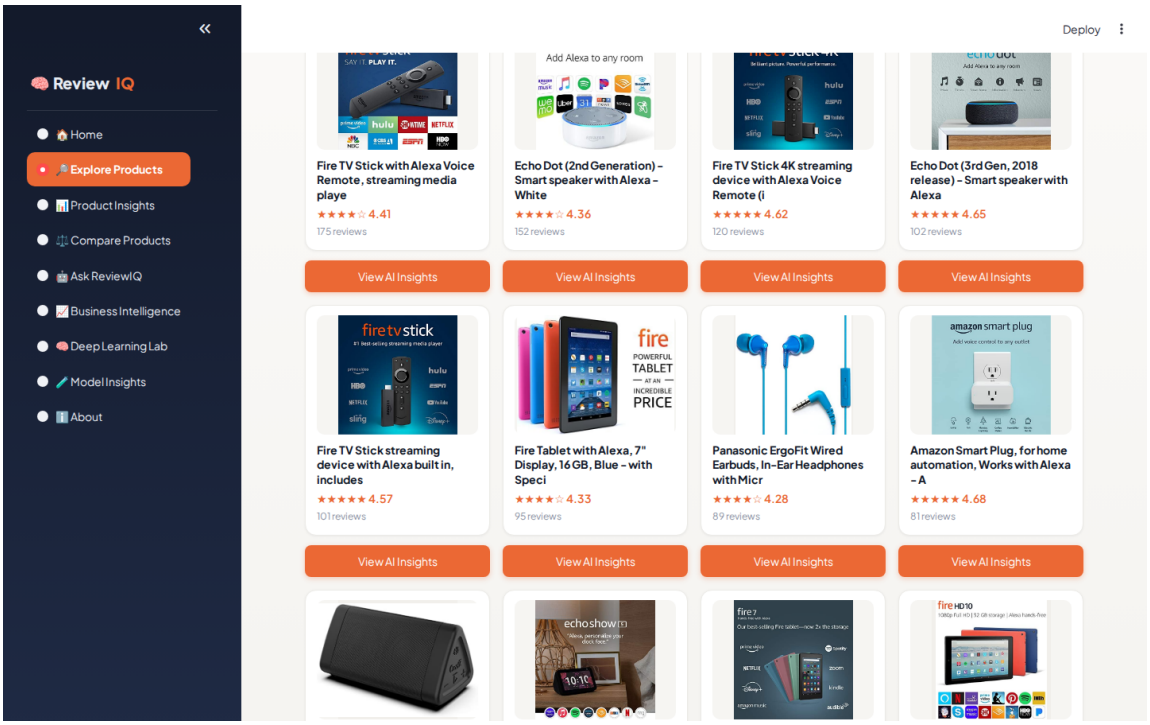


Figure A.3: Explore Products: search and browse by product name.

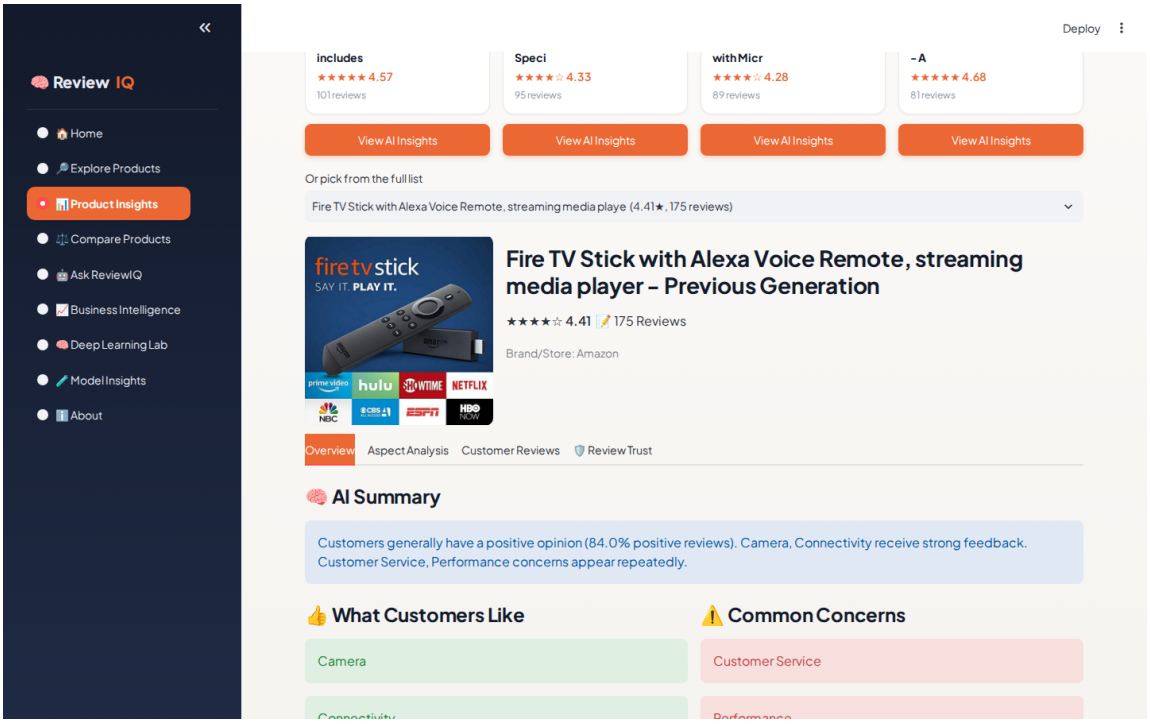


Figure A.4: Product Insights: header, AI summary, strengths and concerns.

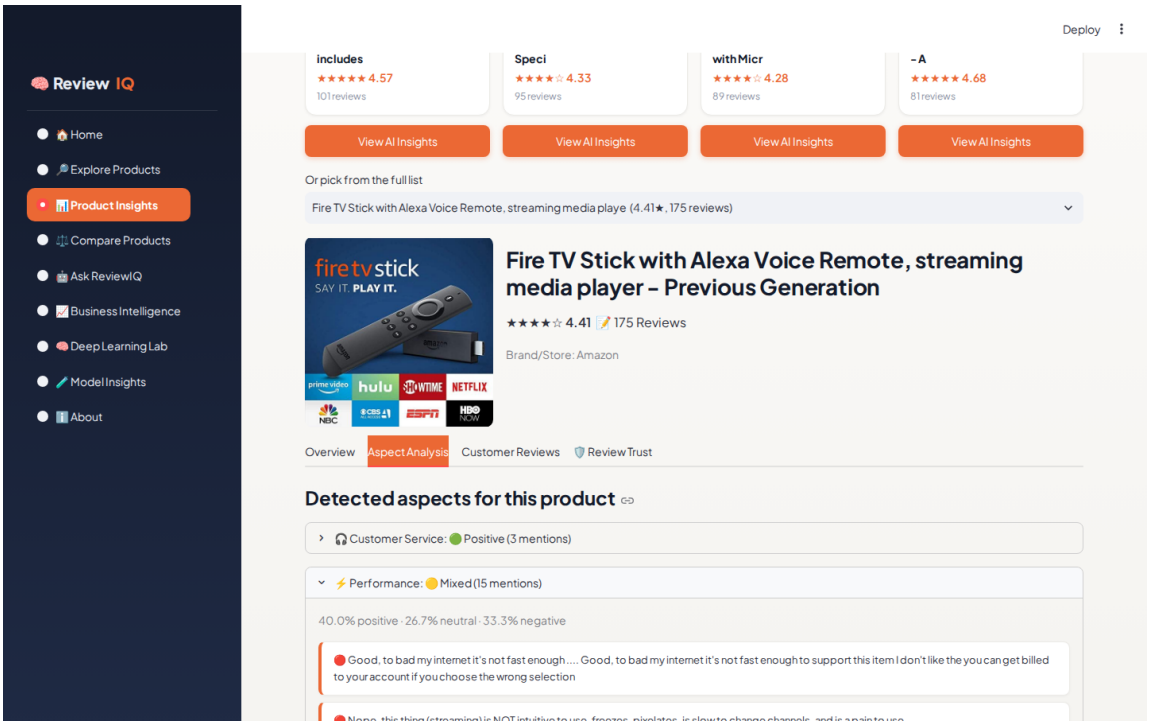


Figure A.5: Product Insights: aspect analysis with supporting evidence sentences.

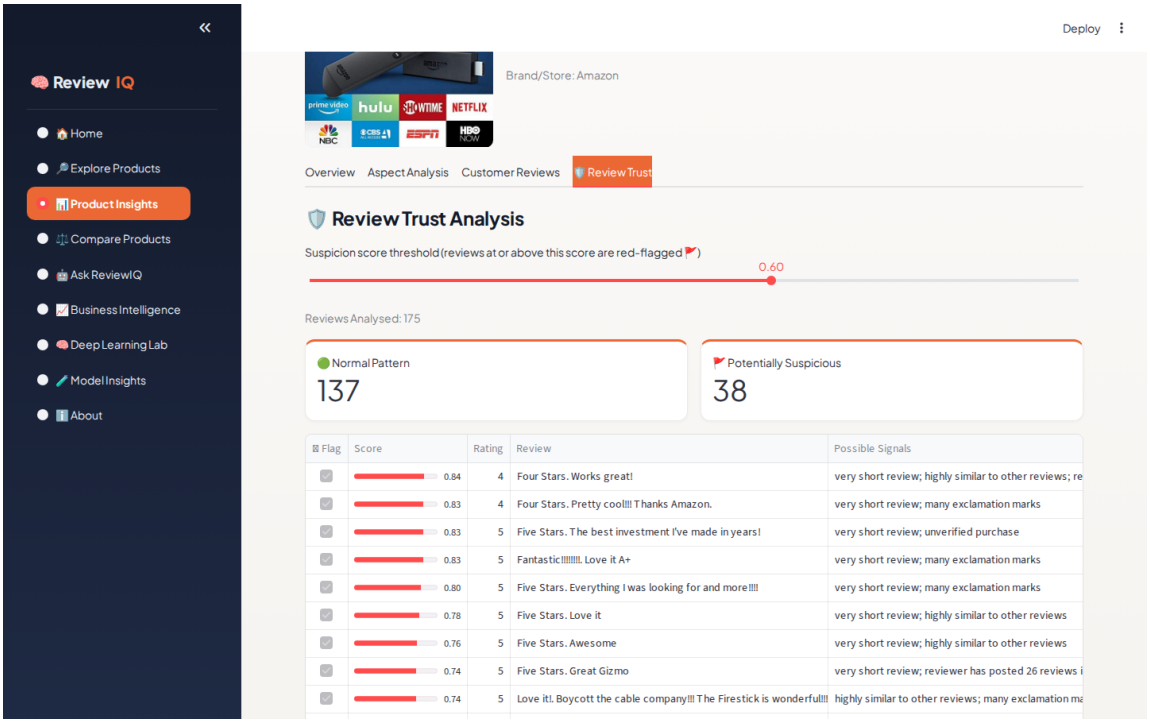


Figure A.6: Product Insights: review trust analysis with threshold slider and flag table.

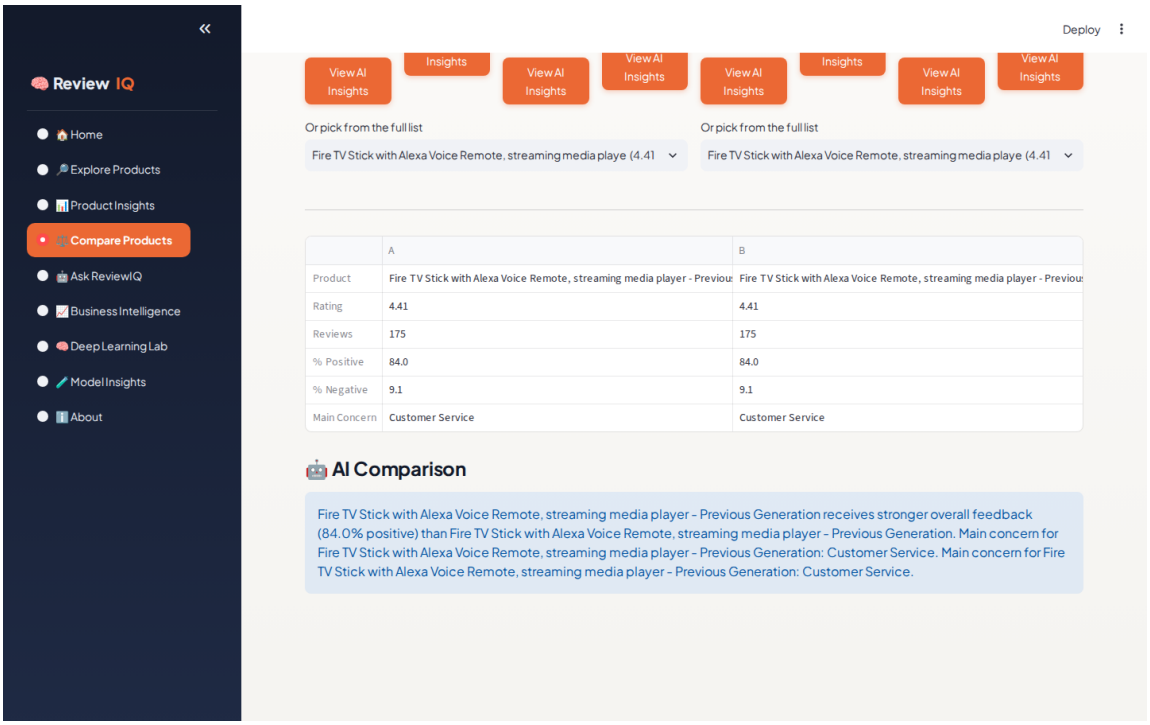


Figure A.7: Compare Products: computed comparison and template-generated summary.

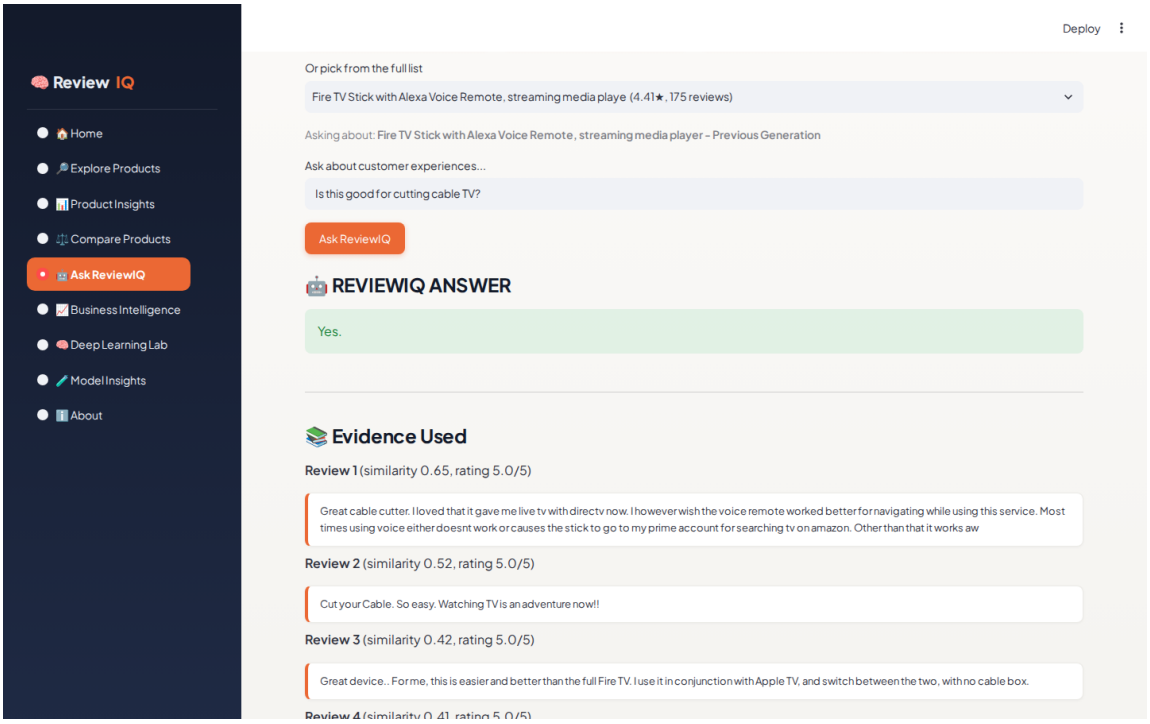


Figure A.8: Ask ReviewIQ: answer with the evidence used.

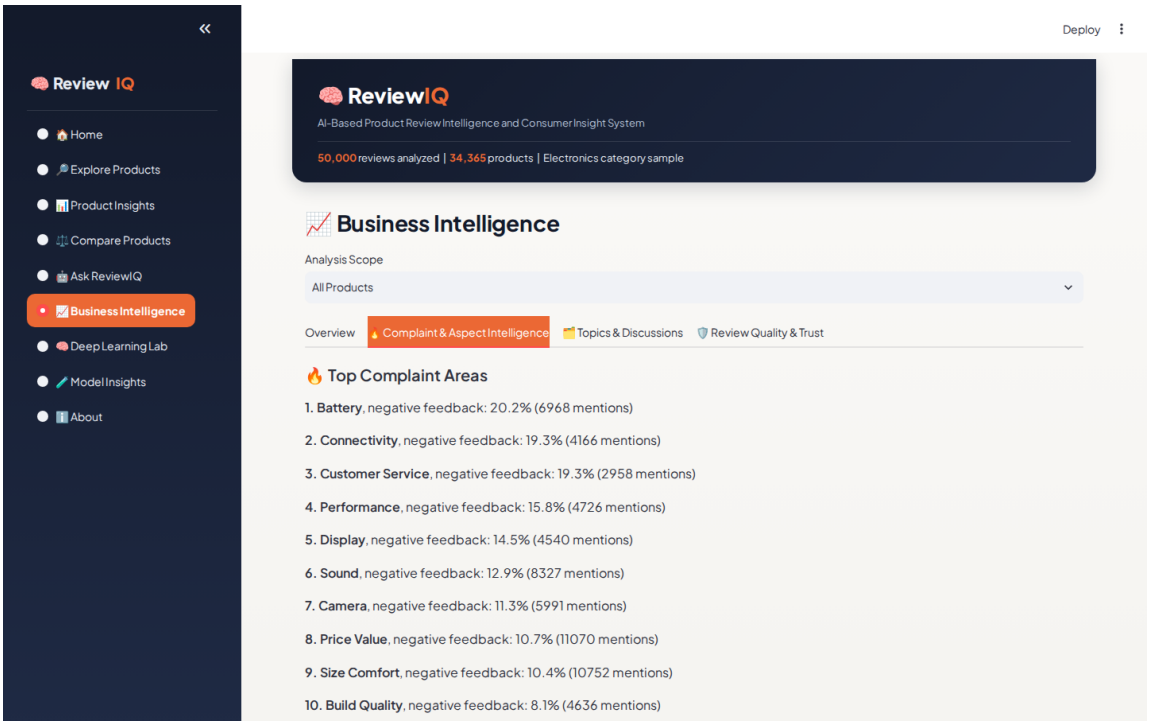


Figure A.9: Business Intelligence: complaint ranking.

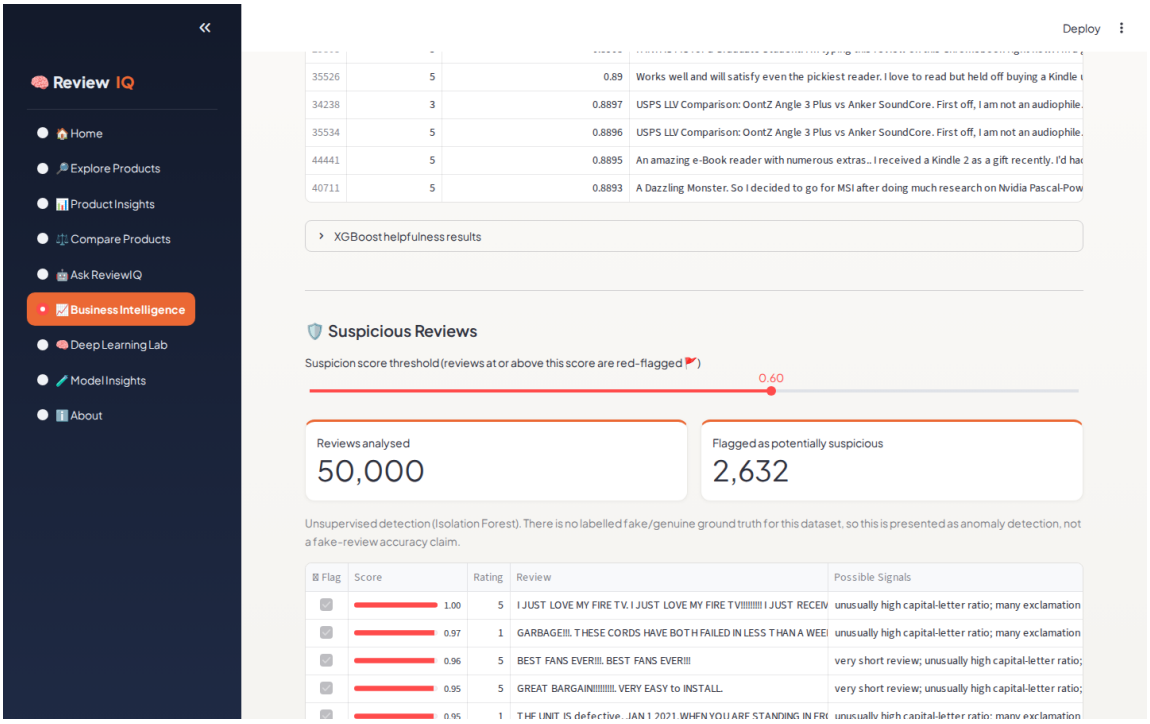


Figure A.10: Business Intelligence: helpfulness and suspicious reviews.

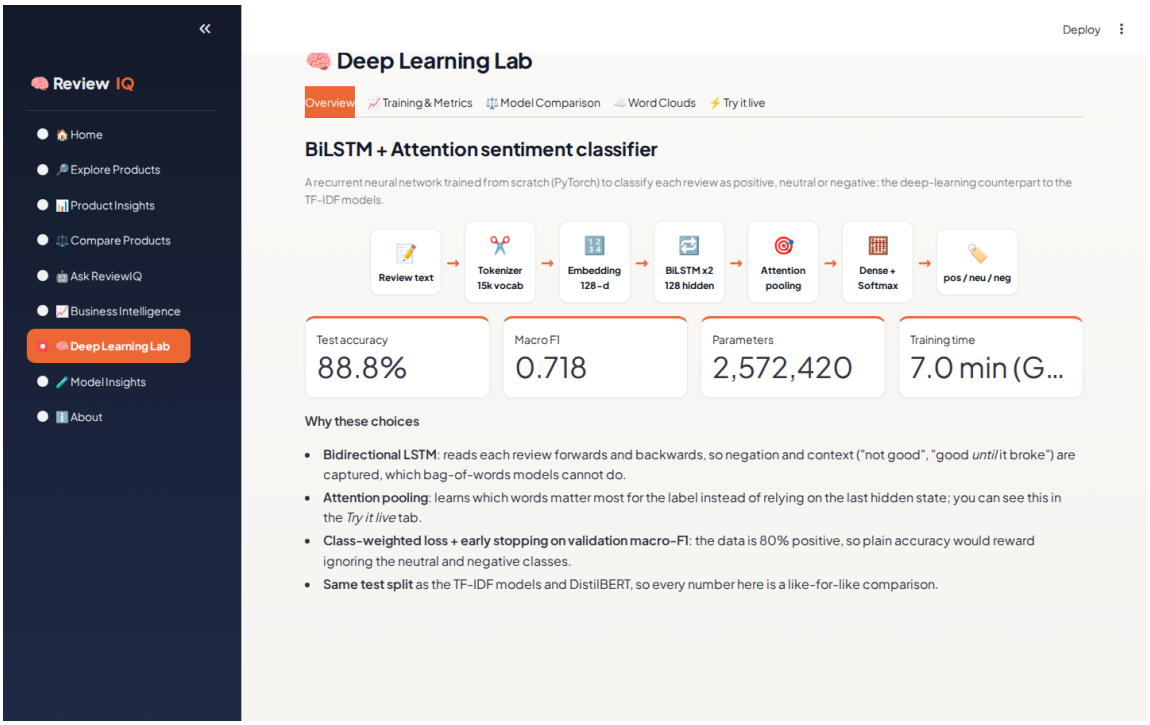


Figure A.11: Deep Learning Lab: overview of the BiLSTM classifier.

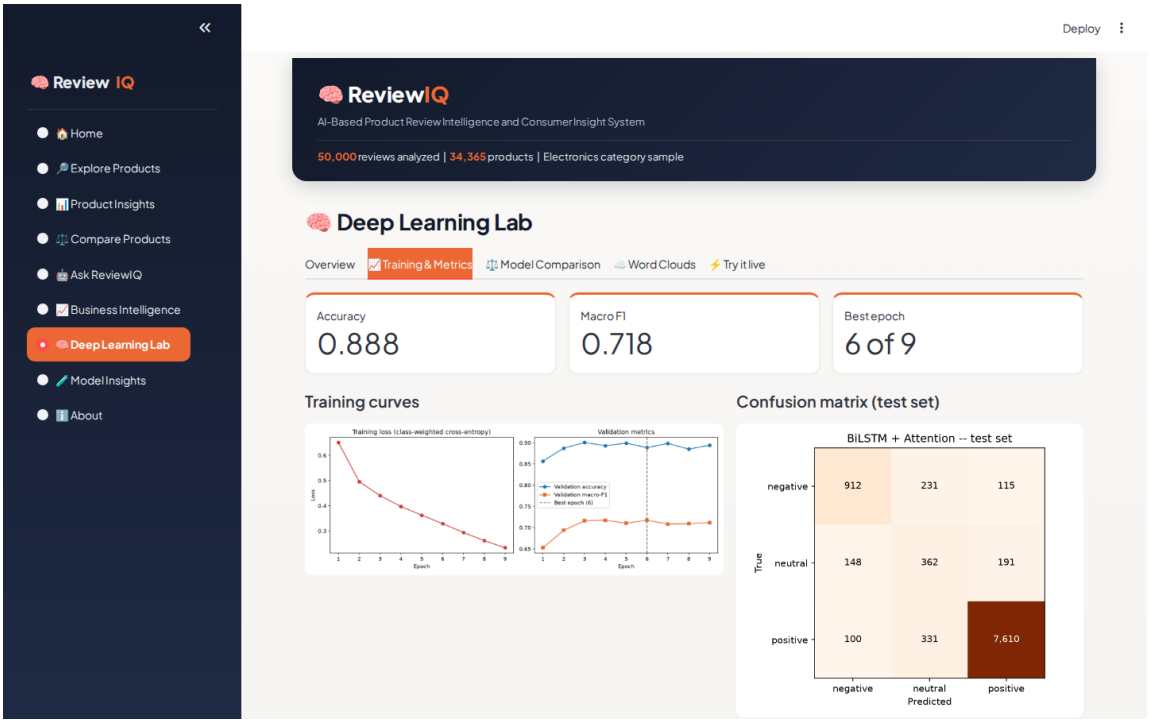


Figure A.12: Deep Learning Lab: training curves, confusion matrix and per-class metrics.

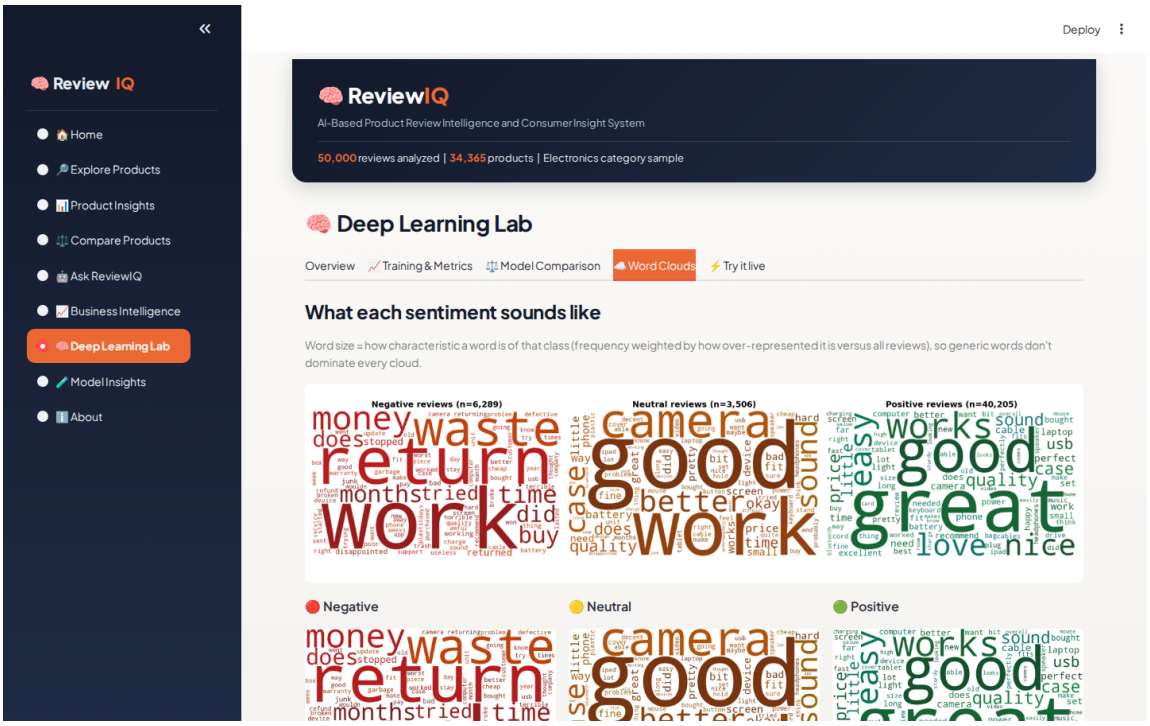


Figure A.13: Deep Learning Lab: word clouds.

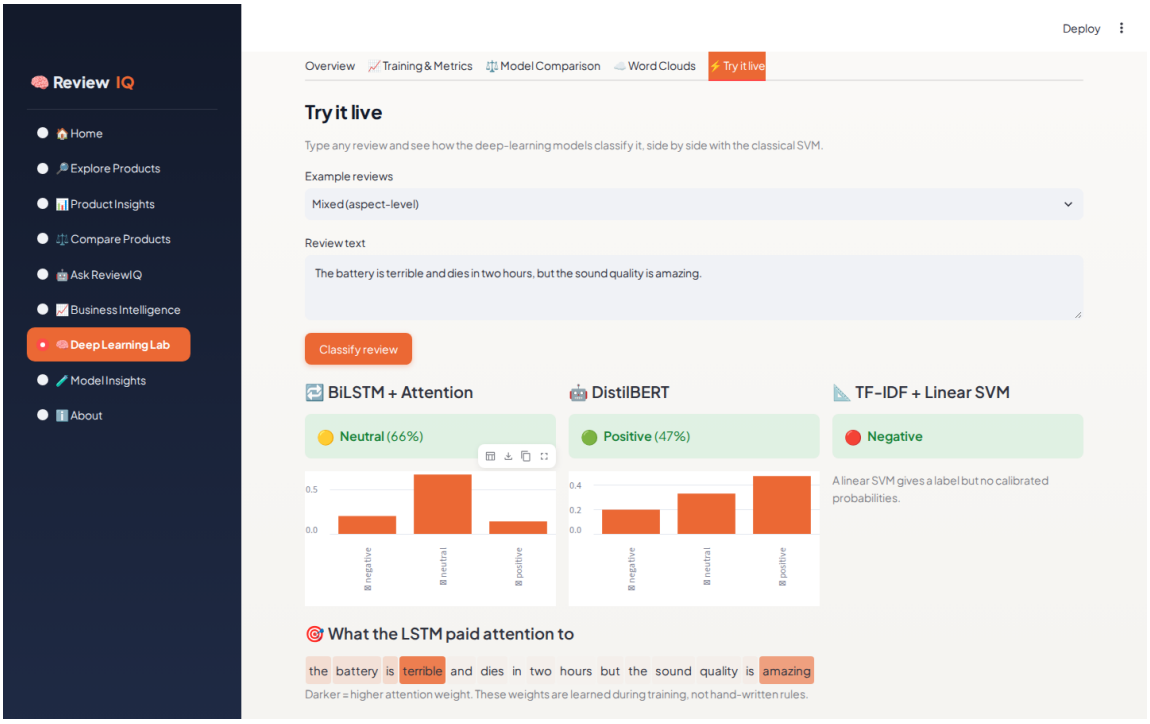


Figure A.14: Deep Learning Lab: live classifier comparing three models, with attention highlighting.

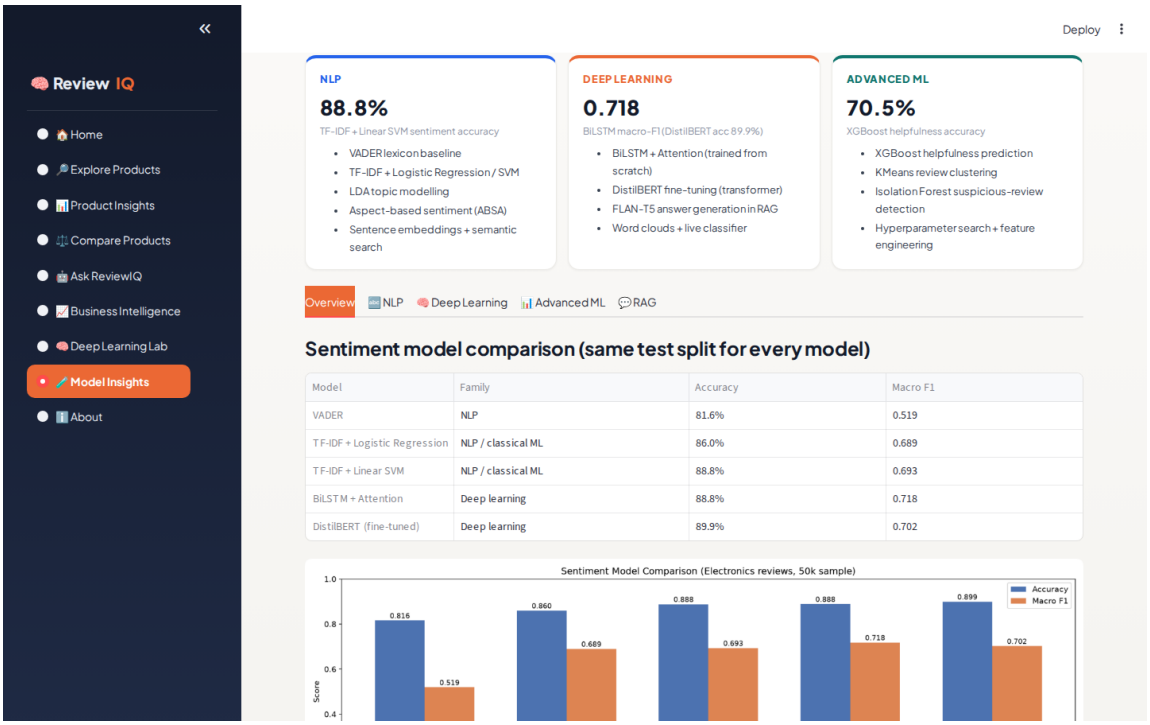


Figure A.15: Model Insights page.

Appendix B

How to Run the Project

1. Create a virtual environment and install the requirements:

```
python3 -m venv .venv
.venv/bin/pip install -r requirements.txt \
    -r requirements-dashboard.txt \
    -r requirements-deep-learning.txt xgboost
```

2. Run the pipeline scripts in order: data preparation, EDA, sentiment models, LDA, ABSA, DistilBERT, XGBoost, KMeans, embeddings, metadata, suspicious reviews, LSTM and word clouds.

3. Start the dashboard and open `http://localhost:8501`:

```
.venv/bin/streamlit run src/dashboard/app.py \
    --server.headless true
```

4. Regenerate the evaluation numbers with these scripts in `src/evaluation`:

```
paper_experiments.py
lstm_ablation.py
inference_cost.py
```

Appendix C

Project Team

Name	Roll number	Role
Nidhi Bhawari	23102A0006	Team member
Aarti Sakpal	23102A0007	Team member
Gayatri Patil	23102A0016	Team member
Prachi Wakode	23102A0020	Team member

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